

Syllabus

CS Seminar: Machine Learning in Practice

Department	Math & CS	Credits	1	Instructor	Sang-Hyun Yoon	Class Room	5708
Subject	Machine Learning in Practice	Class hrs/wk	1	Lab (e-mail)		Attendee	
				5707			

1 Course Objectives

Machine learning is concerned with inductive learning algorithms that automatically generate an algorithm from a set of data. It has a huge potential for real-world algorithmic problems that are ill-defined or intractable such as image classification, object detection, media generation, natural language processing, and PSPACE-hard combinatorial games. This course introduces the basics of machine learning from the practical point of view, and focuses especially on the tools and applications of deep neural networks.

2 Contents

- Introduction to machine learning
- Software toolkit for machine learning: TENSORFLOW/KERAS
- Neural networks
 - perceptrons, activation functions, multi-layer feed-forward networks
 - back-propagation learning, batch normalization, dropout
- Deep neural networks
 - convolution neural networks, convolution/pooling layers
 - transfer learning, residual networks, graph neural networks
- Reinforcement learning
 - Markov decision process, optimal policy, policy/value iteration
 - model-free learning, Monte-Carlo learning, temporal-difference learning
 - SARSA, Q-learning
 - deep reinforcement learning
 - case study: DQN, ALPHAGO
- Generative learning for image processing
 - variational autoencoders, generative adversarial networks
- Natural language processing
 - RNN/LSTM, sequence-to-sequence, attention, Transformers

3 Text & References

Text: None (slides and handouts)

References:

- “*Artificial Intelligence: A Modern Approach*”, S. Russell and P. Norvig
- “*Machine Learning*”, T. Mitchell
- “*Reinforcement Learning*”, S. Sutton and A. Barto
- “*Deep Learning*”, I. Goodfellow, Y. Bengio, and A. Courville
- “*Deep Learning with Python*”, F. Chollet
- “*Generative Deep Learning*”, D. Foster
- “*Natural Language Processing with Transformers*”, L. Tunstall, L. Werra, and T. Wolf
- Course materials on-line:
 - Stanford CS231n: <http://cs231n.stanford.edu>
 - Stanford CS229: <http://cs229.stanford.edu>
 - Stanford CS234: <http://cs234.stanford.edu>
 - Stanford CS20: <http://web.stanford.edu/class/cs20si>
 - MIT AI-related courses: <http://ocw.mit.edu/courses/find-by-topic/#cat=engineering&subcat=computerscience&spec=artificialintelligence>

4 Grading

Activities	Percentages
Assignment	40%
Final Exam	50%
Attendance/Quiz	10%

- Assignments: quadratic/logarithmic regression, image classification, temporal-difference learning, exhaustive search for combinatorial games, α - β pruning, reinforcement learning for 5×5 go
- TENSORFLOW environment will be provided (in COLAB or VMWARE + UBUNTU)
- Final exam (open-book, 3 hrs), no midterm exam
- Make-up final exam for officially approved absence (e.g. COVID, illness, out-of-school competition, conference)
 - immediately after the day all students are released from quarantine/...
 - 0 point if do not show up on the make-up final test

5 Lecture Schedule (Tabular)

#	Topics	Practice/Assignment
1	Course Overview	tool installation
2	Machine Learning Overview	
3	TENSORFLOW Basics	regression, k -clustering
4	Neural Network	digit classifier
5	Convolutional Neural Network	
6-7	Reinforcement Learning	maze, breakout, catch
8	Deep Reinforcement Learning	
9	Combinatorial Games	simple games
10	Search Heuristics for Games	
11	Project: Simple Go	5×5 go
12-13	Case Study: ALPHAGo	
14	VAE & GAN	image generation
15	RNN/LSTM & Transformer	natural language processing
16	Wrapup	

6 Calendar

Mon	Tue	Wed	Thu	Fri
8/28 L1/9	8/29	8/30	8/31	9/1
9/4 L2/3/10	9/5	9/6	9/7	9/8
9/11 L4	9/12	9/13	9/14	9/15
9/18 L5	9/19	9/20	9/21	9/22
9/25 L6	9/26 <i>sports</i>	9/27 <i>holiday</i>	9/28 <i>holiday</i>	9/29 <i>holiday</i>
10/2 <i>holiday</i>	10/3 <i>holiday</i>	10/4	10/5	10/6
10/9 <i>holiday</i>	10/10	10/11	10/12	10/13
10/16 L7	10/17	10/18	10/19	10/20
10/23 L8	10/24 <i>Midt.</i>	10/25 <i>Midt.</i>	10/26 <i>Midt.</i>	10/27 <i>Midt.</i>
10/30 L11	10/31	11/1	11/2	11/3 <i>SAC</i>
11/6 L12	11/7	11/8	11/9	11/10
11/13 L13	11/14	11/15	11/16	11/17
11/20 L14	11/21	11/22	11/23	11/24 <i>SNU</i>
11/27 R	11/28	11/29 <i>KAIST</i>	11/30	12/1
12/4 L15	12/5	12/6	12/7	12/8
12/11 L16	12/12	12/13	12/14	12/15
12/18 <i>Final</i>	12/19 <i>Final</i>	12/20 <i>Final</i>	12/21 <i>Final</i>	12/22 <i>Final</i>